

KAMYAAB

Solution Architecture and MVP Technical Model

Architecture Review Draft v1.0

Purpose: Define the review-ready product and technical architecture for an internal MVP that connects Karachi clients with agent-onboarded blue-collar workers through a client application, web portals, SMS/calls and a tracked booking workflow.

Document status	For team review
Target milestone	Internal MVP usable locally by end of August
Launch coverage	All Karachi supported technically
Primary users	Clients, workers, onboarding agents and operations administrators
Prepared from	Team requirements discussion and the two Kamyab concept documents

Verified Work. Trusted Hands.

Document Map

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Review method: Items marked as a recommendation are proposed architecture decisions. Items marked as open decisions must be confirmed by the team before public launch. The internal MVP may use mocked SMS and payments.

1. Executive Architecture Summary

Kamyaab should be designed as a two-sided local-services marketplace, not as a conventional employment portal or pure worker directory. The client uses a smartphone application, while a worker may participate through an agent-managed profile, SMS/calls or an optional worker web portal. All client-worker contact must be linked to a tracked request or booking so that reviews, complaints and future fees are measurable.

Decision area	Recommended baseline
Product model	Managed directory plus open-job marketplace, using one tracked booking domain.
Client channel	React Native / Expo mobile application.
Worker channel	Agent-managed by default; SMS/call communication; optional phone-authenticated worker portal.
Operations channels	Next.js agent and admin portals.
Backend	TypeScript modular monolith API. No microservices for the MVP.
Data platform	PostgreSQL plus private object storage; Supabase is suitable for the internal MVP.
Matching	Deterministic rules using skill, area, availability, verification and rating.
Voice	Speech-to-text and AI field extraction, followed by client confirmation.
Payments	No real payment integration in internal MVP; free controlled pilot; fixed client booking fee later.
Coverage	All Karachi represented in the data model, with supply density measured by area and category.

Non-negotiable control: Worker phone numbers and exact client addresses should not be published in search results. They should be released only after a tracked request is accepted.

2. Product Definition and Boundaries

2.1 Product statement

Kamyaab helps households and small businesses in Karachi find identity-checked blue-collar workers across approximately 10-12 service categories. Clients browse or post a job through the application. Workers are onboarded in person and may receive opportunities by SMS or call without owning a smartphone.

2.2 What Kamyab is

- **A managed marketplace:** the platform records requests, worker responses, bookings, completion, reviews and complaints.
- **An assisted-access platform:** agents represent workers who cannot maintain a digital profile.
- **A trust layer:** verification checks, review history and operational complaint handling reduce uncertainty.
- **A discovery and dispatch platform:** clients may select a specific worker or submit an open job to eligible workers.

2.3 What Kamyab is not in the MVP

- An employment/payroll platform.
- A guarantee of workmanship, theft prevention or property-damage compensation.
- An escrow or salary-payment platform.
- An AI-first matching engine.
- A multi-city production system.
- A microservices system.

2.4 Initial service categories

Recommended initial categories	Notes
Electrician	High-frequency and urgent household demand.
Plumber	High-frequency and urgent household demand.
Carpenter	Common repair and custom work.
Painter	Project-based work with negotiable scope.
Gardener	Recurring and one-time work.
AC technician	Seasonal but high-urgency service.
Appliance repair technician	Broad domestic demand.
Mason	Project-based repair and construction.
Welder	Specialized household and small-business work.
Cleaner	Short-duration household or office jobs.
Driver	Temporary or contract driving requirements.
General handyman	Small mixed repair tasks.

Safety recommendation: Domestic live-in work, childcare and private security should be excluded from the initial MVP. These categories create substantially higher screening, safeguarding and dispute-resolution requirements.

3. Users, Channels and Trust Model

Actor	Primary channel	Capabilities
Client	Mobile application	Search workers, record voice request, create booking, negotiate price, call after acceptance, review and complain.
Keypad-phone worker	SMS/calls through platform or agent	Receive offers, accept/reject, communicate after acceptance, confirm completion.
Smartphone worker	Optional worker portal	Claim profile, browse open work, accept/reject, update availability, view history.
Onboarding agent	Web portal	Create and maintain worker profiles, inspect CNIC, record checks, service areas and availability.
Operations administrator	Admin portal	Approve/suspend users, manage requests, manually match, resolve complaints, inspect audit logs.
System services	Background jobs and integrations	SMS, call orchestration, AI processing, expiry, reminders and retries.

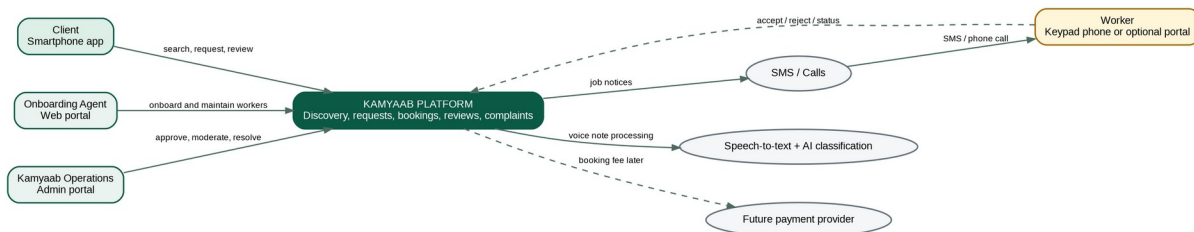


Figure 1 - Kamyab system context

3.1 Verification language

A single “verified” badge is too broad. The UI should show explicit checks so clients understand what was actually validated.

Badge	Meaning
Identity checked	Agent inspected the original CNIC and recorded the document.
Phone confirmed	Worker phone number was confirmed by OTP or call.
Reference checked	At least one reference was contacted and recorded.
Background checked	A documented background-screening process was completed.
Skill assessed	Certificate review, demonstration or agent assessment was recorded.
Kamyaab experience	Worker completed a defined number of tracked bookings.

4. Architecture Principles and Key Decisions

1. Mobile-first for clients; assisted access for workers.
2. One tracked booking workflow for both directory selection and open-job posting.
3. Contact and address privacy until a booking is accepted.
4. Modular monolith before microservices.
5. Rule-based matching before AI matching.
6. Human override for assignment, verification and dispute handling.
7. Private-by-default storage for CNIC, audio and complaint evidence.
8. Every status change creates an auditable history record.
9. External vendors are accessed through adapters so SMS, AI and payments can be replaced.
10. All Karachi is represented by structured areas; operational coverage is measured rather than assumed.

Decision	Rationale	Status
Separate backend API	Required by mobile app, multiple portals and background processing.	Recommended
Express or NestJS modular monolith	Fits a five-person team and short timeline.	Team choice
Supabase PostgreSQL and Storage	Low operational burden for internal MVP.	Recommended
Supabase Auth or equivalent	Supports client phone OTP and staff login.	Recommended
No real telecom dependency locally	Use adapters and mock providers for internal testing.	Recommended
No payment integration in MVP	Payment model is not finalized.	Locked for internal MVP
AI classification requires confirmation	Prevents incorrect automatic job creation.	Locked

5. System Context and Logical Architecture

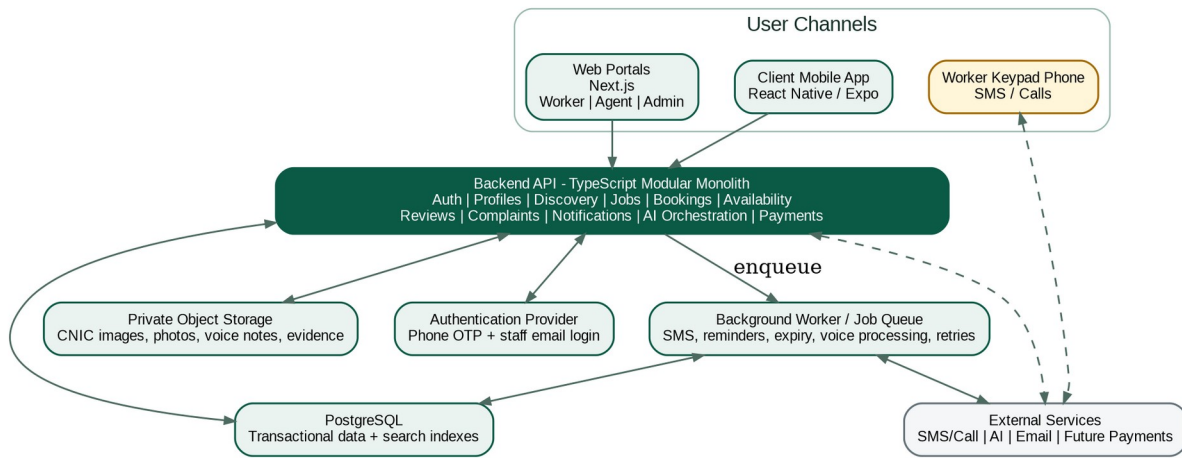


Figure 2 - Logical application architecture

5.1 Technology baseline

Layer	Recommended technology	Purpose
Client application	React Native with Expo	Android-first application, with iOS capability retained.
Web portals	Next.js and TypeScript	Shared worker, agent and admin web interfaces.
Backend API	TypeScript with Express or NestJS	REST API and business rules in a modular monolith.
Database	PostgreSQL	Transactional records, filtering and indexed discovery.
Object storage	Supabase Storage or S3-compatible storage	Private CNIC images, profile photos, voice notes and evidence.
Background work	Database-backed queue or Redis queue	SMS dispatch, AI processing, retries, reminders and expiry.
Authentication	Supabase Auth or equivalent	Phone OTP for clients/workers and staff email authentication.
Testing	Vitest/Jest, API integration tests, Playwright, mobile test tooling	Automated validation across all channels.

6. Core User and Booking Workflows

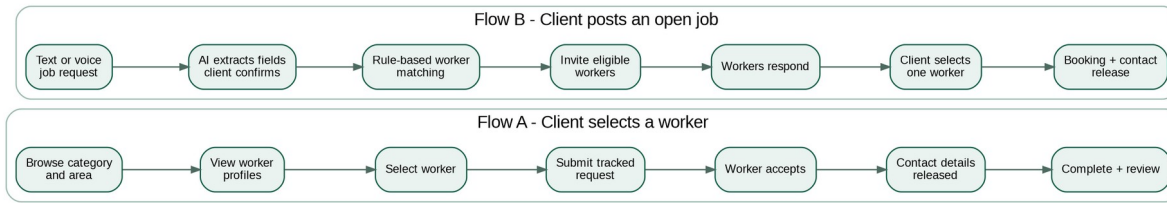


Figure 3 - Two discovery paths converging on one booking model

6.1 Worker onboarding

1. Agent signs in to the agent portal.
2. Agent creates a worker record and captures profile data.
3. Agent inspects the original CNIC and uploads front/back images to private storage.
4. Agent records phone confirmation, address, references, skills, tools, service areas and expected rates.
5. Background or skill checks are recorded as separate verification records.
6. Administrator reviews and approves the worker.
7. Worker becomes searchable only after approval.

6.2 Specific-worker request

1. Client selects category and Karachi area.
2. Search returns approved workers ranked by rules.
3. Client opens a profile and submits a job request to that worker.
4. Worker receives SMS/call or sees the request in the portal.
5. Worker accepts, rejects or lets the request expire.
6. On acceptance, client contact and job address are released to the worker.
7. The parties agree final work price and payment method.
8. Completion is confirmed and the client may review or complain.

6.3 Open-job request

1. Client describes the work by text or voice.
2. AI extracts structured fields and the client confirms them.
3. Matching rules identify eligible workers.
4. A controlled number of workers are invited in batches.
5. Workers accept or reject.
6. Client selects from accepted workers, or operations assigns one manually.
7. A booking is created and contact is released.

Operational safeguard: The platform must always support manual assignment. Automated matching will fail when availability is stale, service areas are incomplete or supply is sparse.

7. Application Modules and API Boundaries

Backend module	Responsibilities
Identity and Access	Authentication mapping, user roles, sessions and account status.
Client Profiles	Client identity, optional email, complaint/no-show flags and address book.
Worker Profiles	Worker data, profile claim, skills, service areas, photos and public summary.
Agents and Verification	Agent ownership, CNIC checks, references, background checks and approval workflow.
Categories and Areas	Configurable service taxonomy and Karachi geographic hierarchy.
Availability	Current state, expiry, update source and history.
Discovery	Filtering and ranking of eligible workers.
Job Requests	Text/voice request data, target worker or open request, budget and urgency.
Matching and Invitations	Eligibility calculation, invitations, response deadlines and batch dispatch.
Bookings	Accepted assignment, price, contact release, status transitions and completion.
Reviews	Booking-gated ratings and comments.
Complaints and Moderation	Complaint intake, evidence, severity, resolution and suspension.
Notifications	SMS, email, in-app events, retry and delivery status.
Voice AI	Audio orchestration, transcript, extraction, confidence and fallback.
Payments	Provider-neutral records for future booking fees and refunds.
Audit	Immutable record of sensitive actions and status transitions.

7.1 API conventions

- Version routes under /api/v1.
- Use JSON request/response contracts validated with schemas such as Zod.
- Return stable machine-readable error codes.
- Authorize each resource by role and ownership, not only by login status.
- Use idempotency keys for booking creation and future payments.
- Use cursor or page-based pagination for listings.

- Do not expose storage object paths directly; issue short-lived signed URLs.

Example endpoint	Purpose
POST /api/v1/job-requests	Create a text or confirmed voice-derived request.
GET /api/v1/workers	Search workers by category, area and availability.
POST /api/v1/job-requests/{id}/invitations	Invite eligible workers.
POST /api/v1/invitations/{id}/accept	Worker accepts a request.
POST /api/v1/bookings/{id}/complete	Confirm completion.
POST /api/v1/bookings/{id}/reviews	Create a booking-gated review.
POST /api/v1/bookings/{id}/complaints	Report a booking issue.
PATCH /api/v1/workers/{id}/availability	Update worker availability.
POST /api/v1/voice-jobs/classify	Transcribe and extract structured fields.

8. Domain and Data Architecture

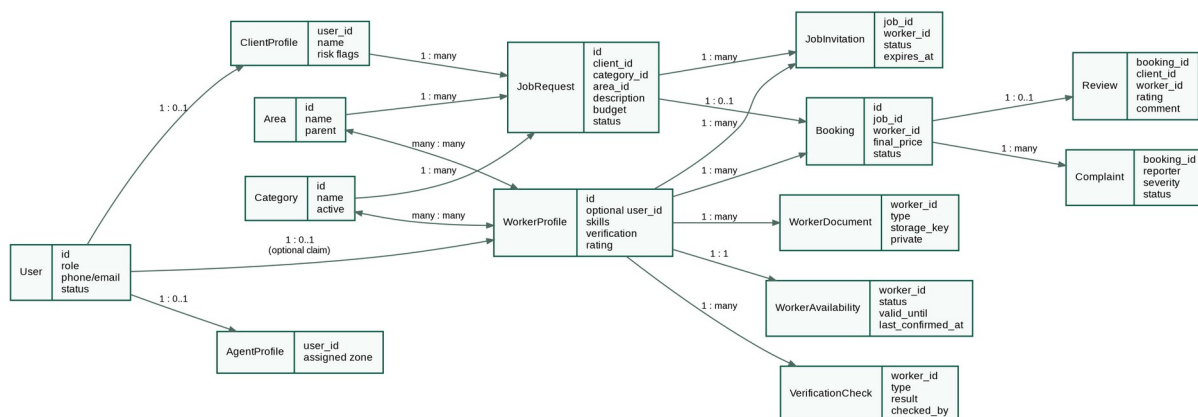


Figure 4 - Simplified domain model

8.1 Core records

Entity group	Key records
Identity	users, roles, client_profiles, agent_profiles, worker_profiles
Verification	worker_documents, worker_references, verification_checks
Catalog	categories, skills, areas, worker_skills, worker_service_areas
Operations	worker_availability, availability_history
Marketplace	job_requests, job_voice_notes, job_matches, job_invitations, job_responses, bookings
Trust	reviews, complaints, complaint_evidence, moderation_actions
Communication	notifications, sms_logs, call_logs
Revenue	payment_records, platform_fees, refunds
Governance	audit_logs, status_history

8.2 Data rules

- Only approved workers are visible in client search.
- Only completed tracked bookings may create public reviews.
- CNIC numbers and images are never returned by public worker-profile APIs.
- A booking references exactly one worker, even if a request invited many workers.
- Every state change is appended to history with actor, reason and timestamp.
- Voice audio, CNIC files and complaint evidence use private storage buckets.
- Worker phone numbers and client addresses are encrypted at rest where supported and masked in logs.

8.3 Karachi area model

Represent Karachi as a configurable hierarchy rather than a free-text address. The MVP can use district/town/area labels without map routing. A worker may serve multiple areas and may optionally define a travel radius later.

Level	Example	Usage
City	Karachi	Top-level tenant/coverage boundary.
District or zone	East, Central, South	Reporting and operational coverage.
Area	Gulshan-e-Iqbal, Clifton, North Nazimabad	Search and service eligibility.
Address	Client-provided exact location	Visible only after acceptance.

9. Authentication and Authorization

Actor	Authentication	Authorization model
Client	Phone OTP; name required; email optional	CLIENT role; own requests, bookings, reviews and complaints only.
Keypad worker	No account required	Agent/admin acts on profile; acceptance may use verified phone interaction.
Smartphone worker	Phone OTP and secure profile-claim flow	WORKER role; own profile, invitations, bookings and availability only.
Agent	Email/password; MFA recommended before live pilot	AGENT role; assigned workers and onboarding operations only.
Administrator	Email/password and MFA	ADMIN or OPS role; moderation and platform operations.
System worker	Service credentials	Minimum service permissions; no interactive login.

9.1 Worker profile claim

1. Agent-created worker record stores a confirmed phone number.
2. Worker signs in with OTP using the same number.
3. System locates the unclaimed worker profile.
4. Additional confirmation is requested, such as CNIC last digits or an agent-issued claim code.
5. Profile is linked to the authenticated user ID.
6. Claim event is written to the audit log.

9.2 Role controls

Sensitive action	Allowed roles
View CNIC documents	Assigned agent, verification operator, administrator
Approve worker	Administrator or verification operator
Change another user's status	Administrator only
View exact client address	Selected worker after acceptance, assigned operations staff
Release worker contact	System after acceptance; operations override is audited
Delete public review	Moderator/administrator with reason
Export user data	Restricted administrator role

10. Worker Availability and Matching

10.1 Availability model

Status	Meaning	Default expiry
AVAILABLE	Worker confirmed ability to take work.	24 hours
BUSY	Worker is engaged until a defined time.	At busy_until
UNAVAILABLE	Worker has explicitly paused work.	Configured date or manual update
UNKNOWN	No recent confirmation.	No expiry; lower search rank

- Smartphone workers update status through the worker portal.
- Keypad workers inform an agent; the agent updates the record and the system sends an SMS confirmation.
- Inbound SMS codes can be added after the internal MVP.
- Availability changes must retain update source: worker portal, SMS, agent, admin or automatic expiry.

10.2 Matching rules

The MVP should use an explainable score. It should first eliminate ineligible workers, then rank the remaining set.

Stage	Rule
Eligibility	Worker approved, category/skill match, service area match, account active.
Priority 1	AVAILABLE workers before UNKNOWN workers.
Priority 2	Recently confirmed availability.
Priority 3	Verification level and safety flags.
Priority 4	Rating and completed Kamyab jobs.
Priority 5	Distance/area preference and recent workload balance.
Fallback	Operations manually assigns or expands the service area.

Do not use AI matching in the MVP: With tens or hundreds of workers, deterministic rules are cheaper, testable and easier to explain when a client or worker challenges a match.

11. Voice and AI Processing



Figure 5 - Voice-note classification pipeline

11.1 Extracted structure

Field	Example
category	Electrician
area	Gulshan-e-Iqbal
description	Ceiling fan is not working
urgency	Today
preferred time	After 5 PM
client budget	Rs. 1,500
materials mentioned	None / fan capacitor

11.2 Controls

- Never submit a job directly from the model output; require client confirmation.
- Store transcript, extracted JSON, model version, confidence and user corrections.
- Apply schema validation and permitted-category validation after AI output.
- Use a low-confidence fallback to manual category and area selection.
- Redact or avoid sending CNIC and unrelated personal data to AI providers.
- Delete voice audio according to a defined retention period after the booking is closed.

12. Notifications, SMS and Calls

Event	Client	Worker	Agent/Admin
Request submitted	In-app confirmation	No notification until invited	Visible in operations queue
Worker invited	Status update	SMS or portal notification	Delivery/failure visible
Worker accepted	Push/in-app/SMS	Confirmation SMS	Booking created
Request expired	In-app/SMS	Expiry SMS optional	Manual fallback queue
Booking reminder	Push/SMS	SMS	Failure alert if undelivered
Completion	Prompt to confirm/review	SMS confirmation	Booking appears in closure queue
Complaint opened	Acknowledgement	Notice where safe	Moderation task

12.1 Provider abstraction

Use an interface such as NotificationProvider with mock, SMS-vendor and future call/IVR implementations. Local development must not depend on a live telecom provider.

- Persist outbound message, provider reference, status, attempts and last error.
- Retry only retryable failures using exponential backoff.
- Prevent duplicate notifications with idempotency keys.
- Mask phone numbers in application logs.
- Provide operations with a failed-delivery queue.

13. Security, Privacy and Safety Controls

Risk	Required control
CNIC exposure	Private bucket, signed URLs, restricted roles, audit reads, retention rules.
Phone/address leakage	Mask in search; release only after acceptance; never log full values.
Fake reviews	Only completed bookings can be reviewed; one client review per booking.
Agent fraud	Agent ownership, approval separation, audit log and periodic verification sampling.
Client harassment	Verified phone, complaint history, suspension and worker reporting.
Worker misconduct	Explicit verification badges, complaint severity, immediate suspension for high-risk reports.
Account takeover	OTP throttling, staff MFA, session revocation and suspicious-login monitoring.
AI data leakage	Send only job content required for classification; no CNIC or stored profile documents.
File abuse	Validate MIME type, extension, size; malware scan before administrative download.
Privilege misuse	Role-based access, least privilege and immutable audit records.

13.1 Platform responsibility

Kamyaab provides	Kamyaab does not initially guarantee
Identity/background-check status	Quality of every job
Tracked booking records	Property-damage compensation
Replacement attempt for no-show/cancellation	Theft compensation
Complaint intake and account suspension	Payment recovery from either party
Review history and evidence trail	Employment benefits or worker insurance

Policy recommendation: For a worker no-show, offer one replacement attempt. Refund the future Kamyaab booking fee when no replacement is available. Do not promise compensation for service quality or property damage until insurance and legal terms exist.

14. Pricing, Payments and Platform Revenue

14.1 Work pricing

1. Platform displays a category-specific suggested starting price or range.
2. Client enters an estimated budget.
3. Worker and client negotiate after acceptance.
4. Final agreed price and payment method may be recorded in the booking.
5. The client pays the worker using any mutually agreed method.

Price disclaimer: Suggested prices are guidance, not guaranteed quotes. Materials, scope, travel and urgency may change the final amount.

14.2 Revenue sequence

Stage	Revenue approach	Architecture impact
Internal MVP	Free; payment records are mocked or optional.	No live payment provider.
Controlled external pilot	Fixed client booking fee before contact release.	Payment adapter and refund record required.
Later marketplace	Percentage commission only if money flows through Kamyab or contractual billing exists.	Provider settlement, ledger, refunds and reconciliation required.

Critical business constraint: A percentage commission cannot be reliably enforced when client and worker negotiate and settle directly in cash. Contact release must be tied to an upfront fixed fee or the complete payment must pass through Kamyab.

15. Deployment, CI/CD and Operations

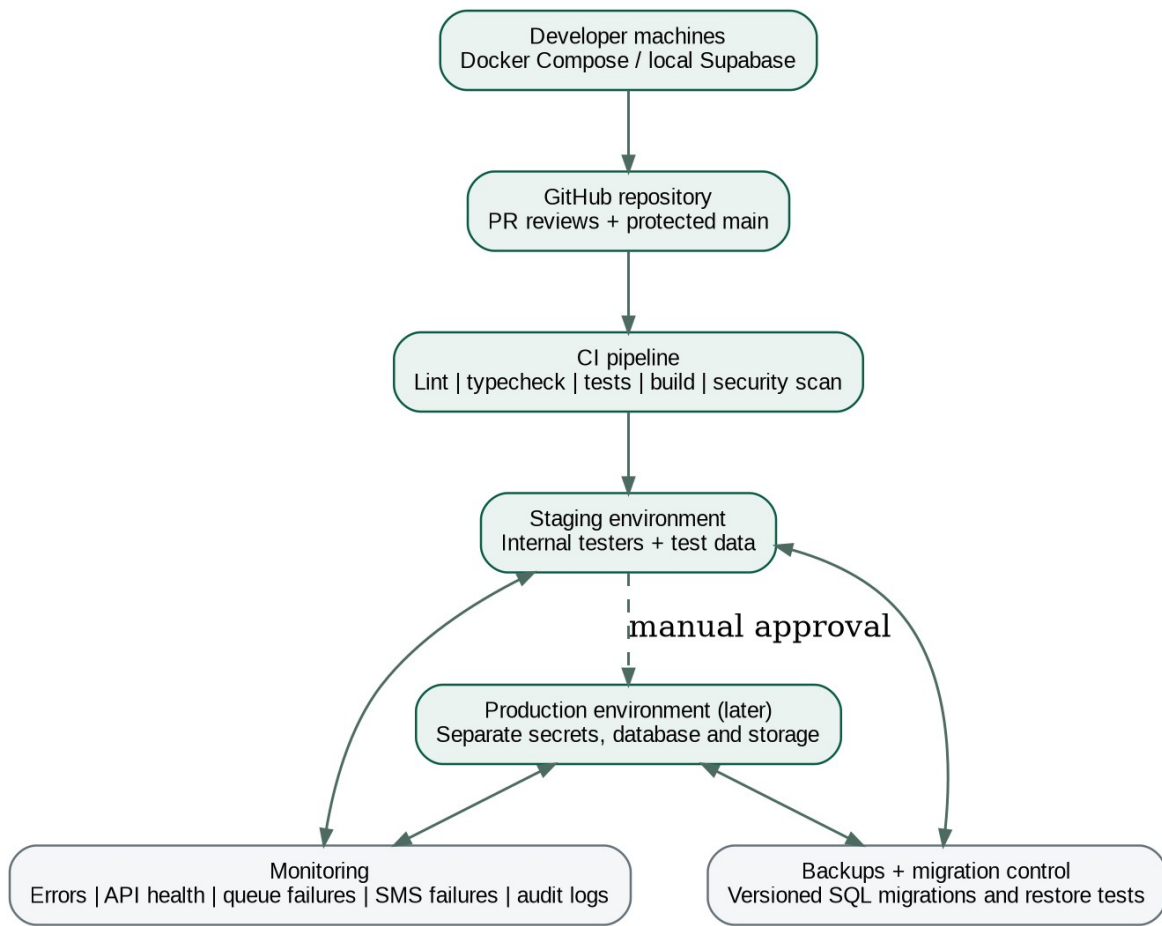


Figure 6 - Delivery and deployment model

15.1 Environments

Environment	Purpose	Data
Local development	Developer feature work and integration through Docker Compose/local services.	Synthetic data only.
Shared staging	Internal testing and demonstrations.	Test workers, clients and mocked communications.
Production later	Controlled pilot and public operations.	Real personal data with separate secrets and storage.

15.2 Repository structure

Area	Suggested repository path
Mobile app	apps/client-mobile
Web portals	apps/portal-web
Backend API	apps/api
Background worker	apps/worker
Shared contracts	packages/contracts
UI/components	packages/ui
Database migrations	database/migrations
Architecture and ADRs	docs/architecture
Infrastructure	infra

15.3 CI pipeline

1. Install dependencies with a locked package manager.
2. Run formatting and lint checks.
3. Run TypeScript type checking.
4. Run unit and API integration tests.
5. Run database migration validation.
6. Build mobile/web/API artifacts.
7. Run dependency and secret scanning.
8. Deploy to staging after merge to main.
9. Run smoke tests.
10. Require manual approval for any later production deployment.

15.4 Operational monitoring

- API error rate, latency and health.
- Background queue depth and failed jobs.
- SMS delivery and retry failures.
- Voice processing failure rate and low-confidence rate.
- Authentication/OTP failures and abuse throttling.
- Booking funnel counts and stalled statuses.
- Database backup success and restoration test results.
- High-severity complaint and suspension events.

16. Testing and Non-Functional Requirements

16.1 Test layers

Layer	Required coverage
Unit	Matching score, status transitions, price validation, availability expiry, permission helpers.
Database	Constraints, migrations, review gating, unique invitation/booking rules and private-data access.
API integration	Authentication, authorization, workflow transitions, file upload and error contracts.
End-to-end web	Agent onboarding, admin approval, worker maintenance and complaint handling.
Mobile integration	Client login, browse, voice request, booking, contact release, completion and review.
Contract	API schema compatibility between mobile, web and backend.
Security	Role isolation, IDOR tests, OTP abuse, file validation and signed URL expiration.
Failure	SMS provider outage, AI timeout, duplicate callback, expired request and queue retry.

16.2 Internal MVP acceptance journeys

1. Agent creates a worker with CNIC, skill and service areas; administrator approves the worker.
2. Client logs in, searches and submits a request to a specific worker.
3. Keypad worker acceptance is simulated through agent/admin or mock SMS.
4. Contact details become visible only after acceptance.
5. Client completes the booking and submits a review.
6. Client records a voice request; AI classifies it; client corrects and submits.
7. Smartphone worker claims a profile and accepts an open job.
8. Administrator manually assigns a worker when automated matching fails.
9. Client opens a complaint and operations suspends an account with an audit reason.

16.3 Non-functional targets for internal MVP

Area	Target
Availability	Best effort for local/internal use; no public SLA.
Performance	Typical API responses under 1 second excluding AI and external providers.
Search	Return the first page of workers under 1 second on test data.
Security	No public access to private documents; role tests must pass.
Recoverability	Versioned migrations; reproducible staging database setup.
Accessibility	Readable labels, large controls and minimal typing in client flows.
Auditability	Sensitive status changes and contact release are recorded.
Maintainability	Modules own business rules; no direct cross-module database writes without service boundaries.

17. Team Delivery Model and Roadmap

17.1 Suggested five-person ownership

Owner	Primary responsibility	Secondary responsibility
Developer 1	Client React Native application	Mobile test automation and UI integration.
Developer 2	Next.js worker/agent/admin portals	Shared UI components and accessibility.
Developer 3	Identity, worker, verification, categories and discovery APIs	Database migrations and seed data.
Developer 4	Jobs, invitations, bookings, notifications and voice AI	Background worker and provider adapters.
Developer 5	QA strategy, CI/CD, integration tests and release coordination	Observability, security tests and documentation.

17.2 Seven-week execution plan

Week	Architecture milestone	Demonstrable outcome
1	Requirements freeze, wireframes, ERD, API contracts and repositories.	Team approves end-to-end flows and ownership.
2	Authentication, roles, categories, areas and worker onboarding.	Agent creates worker; admin approves.
3	Worker discovery, profiles, verification badges and service areas.	Client browses workers by category/area.
4	Job requests, invitations, response and booking state machine.	Specific-worker booking works end to end.
5	Open jobs, worker portal, availability and operations override.	Smartphone worker accepts; admin can manually assign.
6	Voice AI, mock SMS, reviews, complaints and audit trail.	Voice request and trust workflows demonstrated.
7	Integration, security testing, bug fixing, local deployment and team handover.	Internal MVP acceptance journeys pass.

18. Risks, Assumptions and Review Decisions

Risk	Why it matters	Mitigation
All-Karachi launch with thin supply	Clients may see workers who are too far away or unavailable.	Measure coverage by category/area; rank by service area; permit manual expansion.
Stale worker availability	Directory results become unreliable.	24-hour expiry, UNKNOWN status and agent refresh workflow.
Commission bypass	Direct contact allows off-platform settlement.	Use tracked requests; future fixed fee before contact release.
Verification overclaim	CNIC inspection may be misunderstood as skill/safety guarantee.	Use separate badges and explicit definitions.
Agent fraud or poor data	Profiles and checks depend on agent quality.	Approval separation, audit logs, sampling and agent performance metrics.
AI classification errors	Wrong category/area produces bad matches.	Confidence threshold, validation and client confirmation.
Telecom dependency	SMS delivery may fail or be expensive.	Provider adapter, mock local provider, failure queue and agent-call fallback.
Scope overload	Mobile, multiple portals, AI and 12 categories may exceed deadline.	Reuse shared contracts, keep payments mocked, exclude advanced AI and public launch.
Sensitive-data breach	CNIC and addresses create high impact.	Private storage, least privilege, signed URLs and retention controls.
Dispute expectations	Users may assume Kamyab guarantees work or compensation.	Clear terms, limited replacement policy and documented complaint procedure.

18.1 Decisions to confirm during team review

Decision	Recommended default	Owner / deadline
Backend framework	NestJS if team accepts structure; Express if team prefers minimal framework.	Engineering lead - Week 1
Authentication provider	Supabase Auth for MVP.	Backend team - Week 1
Background-check definition	Document exact checks and badge text.	Product/operations - Week 1
Initial area taxonomy	Agree Karachi district/area seed list.	Product - Week 1
Minimum-price source	Admin-configured guidance by category.	Product/operations - Week 2
SMS/call vendor	Mock internally; evaluate local providers before pilot.	Operations/engineering - Post-MVP
Background queue	Database-backed queue first or Redis/BullMQ based on hosting.	Backend team - Week 2
Voice providers	Select speech-to-text and LLM based on Urdu quality and cost.	AI owner - Week 3
Data retention	CNIC, voice and complaint evidence retention periods.	Product/legal - Before pilot
Fee model	Fixed booking fee before contact release.	Founders - Before external pilot

Appendix A. State Models and MVP Scope

A.1 Job request state model

State	Meaning	Possible next states
DRAFT	Client has not submitted.	SUBMITTED, CANCELLED
SUBMITTED	Request is accepted by the platform.	MATCHING, CANCELLED
MATCHING	Eligibility and invitation process active.	WORKER_CONTACTED, EXPIRED, CANCELLED
WORKER_CONTACTED	At least one worker has been invited.	ACCEPTED, EXPIRED, CANCELLED
ACCEPTED	One worker accepted and was selected.	CONTACT_SHARED, CANCELLED
CONTACT_SHARED	Both sides may communicate.	IN_PROGRESS, CANCELLED, NO_SHOW
IN_PROGRESS	Work has started.	COMPLETED, DISPUTED, CANCELLED
COMPLETED	Work closure confirmed.	DISPUTED
EXPIRED	No worker accepted in time.	MATCHING through manual retry
NO_SHOW	Assigned worker did not arrive.	MATCHING, CANCELLED
DISPUTED	Complaint requires operations action.	COMPLETED, CANCELLED

A.2 Invitation state model

State	Meaning
PENDING	Created but not yet dispatched.
SENT	Provider accepted the message.
DELIVERED	Provider confirmed delivery where available.
ACCEPTED	Worker accepted before expiry.
REJECTED	Worker rejected.
EXPIRED	No response before deadline.
FAILED	Delivery or processing failed.

A.3 Internal MVP - in scope

- Client phone login and worker browsing.
- 12 configurable worker categories and Karachi areas.
- Agent onboarding with private CNIC upload.
- Admin approval, suspension and manual assignment.
- Specific-worker and open-job requests.
- Optional worker profile claim and worker portal.
- Rule-based matching and expiring availability.
- Voice recording, AI classification and client confirmation.
- Mock SMS/call events.
- Booking state history, review and complaint handling.
- Local/staging deployment with automated tests.

A.4 Internal MVP - explicitly out of scope

- Public production launch.
- Real percentage commission or escrow.
- Automated police/NADRA integration.
- Call masking or IVR short code.
- Insurance and damage guarantees.
- AI worker-ranking model.
- Materials/supplier marketplace.
- Housing-society B2B dashboard.
- Multi-city operations.

A.5 Source inputs

This architecture draft consolidates and resolves product assumptions from:

- Kamyab - AI Dot (Straw Hats).pdf
- Kamyab_first draft.pdf
- Team answers captured during the architecture discovery discussion.

Architecture Review Sign-off

Reviewer	Role	Decision	Comments / required changes
		Approve / Approve with changes / Reject	
		Approve / Approve with changes / Reject	
		Approve / Approve with changes / Reject	

Recommended approval condition: Approve the architecture baseline after the team confirms the verification definition, Karachi area taxonomy, backend framework, queue choice and fee model. These decisions do not block the first two weeks of internal MVP development if interfaces remain provider-neutral.